

TERM PROJECT 1

CODE:

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1 from LinkBinaryTree import LinkBinaryTree as Tree
2
3 # Equation 1
4 tree = Tree()
5 print("Equation 1: (3 * 5) - ((4 * 5) + (6 - 7))")
6 tree._add_root('-')
7 tree._add_left(tree.root(), e: '*')
8 tree._add_right(tree.root(), e: '+')
9 tree._add_left(tree.left(tree.root()), e: 3)
10 tree._add_right(tree.left(tree.root()), e: 5)
11 tree._add_left(tree.right(tree.root()), e: '*')
12 tree._add_right(tree.right(tree.root()), e: '-')
13 tree._add_left(tree.left(tree.right(tree.root())), e: 4)
14 tree._add_right(tree.left(tree.right(tree.root())), e: 5)
15 tree._add_left(tree.right(tree.right(tree.root())), e: 6)
16 tree._add_right(tree.right(tree.right(tree.root())), e: 7)
17
18 print("Preorder traversal: ", end=" ")
19 for i in tree.positions():
20     print(i.element(), end=" ")
21 print()
22
23 print("Postorder traversal: ", end=" ")
24 for i in tree.postorder():
25     print(i.element(), end=" ")
26 print()
27
28 print("Inorder traversal: ", end=" ")
29 for i in tree.inorder():
30     print(i.element(), end=" ")
31 print()
32 print()
33
34 # Equation 2
35 tree = Tree()
36 print("Equation 2: ((a + b) * c) - (d - e)")
37 tree._add_root('-')
38 tree._add_left(tree.root(), e: '*')
39 tree._add_right(tree.root(), e: '-')
40 tree._add_left(tree.left(tree.root()), e: '+')
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40 tree._add_left(tree.left(tree.root()), e: '+')
41 tree._add_right(tree.left(tree.root()), e: 'c')
42 tree._add_left(tree.right(tree.root()), e: 'd')
43 tree._add_right(tree.right(tree.root()), e: 'e')
44 tree._add_left(tree.left(tree.left(tree.root())), e: 'a')
45 tree._add_right(tree.left(tree.left(tree.root())), e: 'b')
46
47 print("Preorder traversal: ", end=" ")
48 for i in tree.positions():
49     print(i.element(), end=" ")
50 print()
51
52 print("Postorder traversal: ", end=" ")
53 for i in tree.postorder():
54     print(i.element(), end=" ")
55 print()
56
57 print("Inorder traversal: ", end=" ")
58 for i in tree.inorder():
59     print(i.element(), end=" ")
60 print()
61 print()
62
63 # Equation 3
64 tree = Tree()
65 print("Equation 3: ((a ^ b) + (c + d)) + ((e * f) / (g + h))")
66 tree._add_root('+')
67 tree._add_left(tree.root(), e: '+')
68 tree._add_right(tree.root(), e: '/')
69 tree._add_left(tree.left(tree.root()), e: '^')
70 tree._add_right(tree.left(tree.root()), e: '+')
71 tree._add_left(tree.right(tree.root()), e: '*')
72 tree._add_right(tree.right(tree.root()), e: '+')
73 tree._add_left(tree.left(tree.left(tree.root())), e: 'a')
74 tree._add_right(tree.left(tree.left(tree.root())), e: 'b')
75 tree._add_left(tree.right(tree.left(tree.root())), e: 'c')
76 tree._add_right(tree.right(tree.left(tree.root())), e: 'd')
77 tree._add_left(tree.left(tree.right(tree.root())), e: 'e')
78 tree._add_right(tree.left(tree.right(tree.root())), e: 'f')
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79 tree._add_left(tree.right(tree.right(tree.root())), e: 'g')
80 tree._add_right(tree.right(tree.right(tree.root())), e: 'h')
81
82 print("Preorder traversal: ", end=" ")
83 for i in tree.positions():
84     print(i.element(), end=" ")
85 print()
86
87 print("Postorder traversal: ", end=" ")
88 for i in tree.postorder():
89     print(i.element(), end=" ")
90 print()
91
92 print("Inorder traversal: ", end=" ")
93 for i in tree.inorder():
94     print(i.element(), end=" ")
95 print()
96 print()
97
98 # Equation 4
99 tree = Tree()
100 print("Equation 4: (a + b) / (c * (d - (e ^ f)))")
101 tree._add_root('/')
102 tree._add_left(tree.root(), e: '+')
103 tree._add_right(tree.root(), e: '*')
104 tree._add_left(tree.left(tree.root()), e: 'a')
105 tree._add_right(tree.left(tree.root()), e: 'b')
106 tree._add_left(tree.right(tree.root()), e: 'c')
107 tree._add_right(tree.right(tree.root()), e: '-')
108 tree._add_left(tree.right(tree.right(tree.root())), e: 'd')
109 tree._add_right(tree.right(tree.right(tree.root())), e: '^')
110 tree._add_left(tree.right(tree.right(tree.right(tree.root()))), e: 'e')
111 tree._add_right(tree.right(tree.right(tree.right(tree.root()))), e: 'f')
112
113 print("Preorder traversal: ", end=" ")
114 for i in tree.positions():
115     print(i.element(), end=" ")
116 print()

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156 print("Inorder traversal: ", end=" ")
157 for i in tree.inorder():
158     print(i.element(), end=" ")
159 print()
160 print()
161
162 # Equation 6
163 tree = Tree()
164 print("Equation 6: (((5 + 2) * (2 - 1)) / ((2 + 9) + ((7 - 2) - 1))) * 8")
165 tree._add_root('*')
166 tree._add_left(tree.root(), e: '/')
167 tree._add_right(tree.root(), e: '8')
168 tree._add_left(tree.left(tree.root()), e: '*')
169 tree._add_right(tree.left(tree.root()), e: '+')
170 tree._add_left(tree.left(tree.left(tree.root()))), e: '+'
171 tree._add_right(tree.left(tree.left(tree.root()))), e: '-'
172 tree._add_left(tree.right(tree.left(tree.root()))), e: '+'
173 tree._add_right(tree.right(tree.left(tree.root()))), e: '-'
174 tree._add_left(tree.left(tree.left(tree.left(tree.root()))), e: '5')
175 tree._add_right(tree.left(tree.left(tree.left(tree.root()))), e: '2')
176 tree._add_left(tree.right(tree.left(tree.left(tree.root()))), e: '2')
177 tree._add_right(tree.right(tree.left(tree.left(tree.root()))), e: '1')
178 tree._add_left(tree.left(tree.right(tree.left(tree.root()))), e: '2')
179 tree._add_right(tree.left(tree.right(tree.left(tree.root()))), e: '9')
180 tree._add_left(tree.right(tree.right(tree.left(tree.root()))), e: '-')
181 tree._add_right(tree.right(tree.right(tree.left(tree.root()))), e: '1')
182 tree._add_left(tree.left(tree.right(tree.right(tree.left(tree.root()))), e: '7')
183 tree._add_right(tree.left(tree.right(tree.right(tree.left(tree.root()))), e: '2')
184
185 print("Preorder traversal: ", end=" ")
186 for i in tree.positions():
187     print(i.element(), end=" ")
188 print()
189
190 print("Postorder traversal: ", end=" ")
191 for i in tree.postorder():
192     print(i.element(), end=" ")
193 print()

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118 print("Postorder traversal: ", end=" ")
119 for i in tree.postorder():
120     print(i.element(), end=" ")
121 print()
122
123 print("Inorder traversal: ", end=" ")
124 for i in tree.inorder():
125     print(i.element(), end=" ")
126 print()
127 print()
128
129 # Equation 5
130 tree = Tree()
131 print("Equation 5: ((a - b) + c) * ((d + e) * (f / g))")
132 tree._add_root('*')
133 tree._add_left(tree.root(), e: '+')
134 tree._add_right(tree.root(), e: '*')
135 tree._add_left(tree.left(tree.root()), e: '-')
136 tree._add_right(tree.left(tree.root()), e: 'c')
137 tree._add_left(tree.right(tree.root()), e: '+')
138 tree._add_right(tree.right(tree.root()), e: '/')
139 tree._add_left(tree.left(tree.left(tree.root()))), e: 'a')
140 tree._add_right(tree.left(tree.left(tree.root()))), e: 'b')
141 tree._add_left(tree.left(tree.right(tree.root()))), e: 'd')
142 tree._add_right(tree.left(tree.right(tree.root()))), e: 'e')
143 tree._add_left(tree.right(tree.right(tree.root()))), e: 'f')
144 tree._add_right(tree.right(tree.right(tree.root()))), e: 'g')
145
146 print("Preorder traversal: ", end=" ")
147 for i in tree.positions():
148     print(i.element(), end=" ")
149 print()
150
151 print("Postorder traversal: ", end=" ")
152 for i in tree.postorder():
153     print(i.element(), end=" ")
154 print()

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195 print("Inorder traversal: ", end=" ")
196 for i in tree.inorder():
197     print(i.element(), end=" ")
198 print()
199 print()

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OUTPUT:

Equation 1: $(3 * 5) - ((4 * 5) + (6 - 7))$

Preorder traversal: $- * 3 5 + * 4 5 - 6 7$

Postorder traversal: $3 5 * 4 5 * 6 7 - + -$

Inorder traversal: $3 * 5 - 4 * 5 + 6 - 7$

Equation 2: $((a + b) * c) - (d - e)$

Preorder traversal: $- * + a b c - d e$

Postorder traversal: $a b + c * d e - -$

Inorder traversal: $a + b * c - d - e$

Equation 3: $((a ^ b) + (c + d)) + ((e * f) / (g + h))$

Preorder traversal: $+ + ^ a b + c d / * e f + g h$

Postorder traversal: $a b ^ c d + + e f * g h + / +$

Inorder traversal: $a ^ b + c + d + e * f / g + h$

Equation 4: $(a + b) / (c * (d - (e ^ f)))$

Preorder traversal: $/ + a b * c - d ^ e f$

Postorder traversal: $a b + c d e f ^ - * /$

Inorder traversal: $a + b / c * d - e ^ f$

Equation 5: $((a - b) + c) * ((d + e) * (f / g))$

Preorder traversal: $* + - a b c * + d e / f g$

Postorder traversal: $a b - c + d e + f g / * *$

Inorder traversal: $a - b + c * d + e * f / g$

Equation 6: $((((5 + 2) * (2 - 1)) / ((2 + 9) + ((7 - 2) - 1))) * 8$

Preorder traversal: $* / * + 5 2 - 2 1 + + 2 9 - - 7 2 1 8$

Postorder traversal: $5 2 + 2 1 - * 2 9 + 7 2 - 1 - + / 8 *$

Inorder traversal: $5 + 2 * 2 - 1 / 2 + 9 + 7 - 2 - 1 * 8$